a language like C or C++ that lacks memory safety, then failing to check this assumption can lead to a serious class of attacks known as *buffer overflow* attacks. Even in a memory-safe language, failing to check that inputs to an API match the developer's assumptions can result in unwanted behavior.

> **DEFINITION** A ***buffer overflow*** or ***buffer overrun*** occurs when an attacker can supply input that exceeds the size of the memory region allocated to hold that input. If the program, or the language runtime, fails to check this case then the attacker may be able to overwrite adjacent memory.

A buffer overflow might seem harmless enough; it just corrupts some memory, so maybe we get an invalid value in a variable, right? However, the memory that is overwritten may not always be simple data and, in some cases, that memory may be interpreted as code, resulting in a *remote code execution* vulnerability. Such vulnerabilities are extremely serious, as the attacker can usually then run code in your process with the full permissions of your legitimate code.

> **DEFINITION** ***Remote code execution*** (RCE) occurs when an attacker can inject code into a remotely running API and cause it to execute. This can allow the attacker to perform actions that they would not normally be allowed.

In the Natter API code, the input to the API call is presented as structured JSON. As Java is a memory-safe language, you don't need to worry too much about buffer overflow attacks.